07-30 第2篇

阅读理解（说明文） Why Do We Have Fingerprints?

① Look at your own thumb under a lamp for a moment. You will see tiny, curved ridges that form loops, whorls, or arches. These are your **fingerprints**, and even if you scrub them hard with soap and water every day for twenty years, they will never wash off. They will be the same shape when you are old as they were when you were born. In fact, of the 8 billion people alive on Earth today, **not two of them share exactly the same fingerprints** — not even identical twins. But why do we have them at all? Nature rarely builds something for no reason. Scientists have been trying to answer this simple-looking question for more than a hundred years, and today they believe fingerprints do several quiet jobs at the same time.

② For a long time, the most popular answer was that fingerprints help us **grip**. If you try to hold a glass with wet fingers, you can often feel the grooves pressing into the surface. This seemed like a neat explanation. But in 2009, researchers from the University of Manchester tested the idea with care. They asked people to pull smooth objects through their fingers, and carefully measured the friction — the force that stops things from slipping. Surprisingly, flat, smooth fingers gripped just as well as ridged ones, and sometimes a little better. The "grip" idea, it turned out, was only part of the story.

③ A more recent idea is that fingerprints may help us **handle wet surfaces**. When we touch a wet object, the ridges allow extra water to escape down the tiny valleys between them, similar to the grooves on a car tyre that push water away in the rain. On a dry object, this does not matter much, but on a rainy day — or when we are washing fruit — the ridges help keep our skin in direct contact with the surface. Simple tests with wet and dry objects have supported this idea.

④ A second idea is that fingerprints make our fingers **more sensitive**. Under the ridges lie groups of tiny nerves that notice tiny shakes of the skin. When we slide a finger over a surface, the ridges shake at a certain speed against the small bumps of the object. This shaking tells the brain what the surface feels like — silk, sand, paper, glass. A completely smooth finger gives the brain fewer signals. In other words, fingerprints may be a natural "microphone" for touch.

⑤ A third, very different idea is that fingerprints **protect our skin from small damage**. The ridges rise and fall slightly as we use our hands, which keeps the skin flexible. Without them, small cuts and tiny tears might appear more often. Scientists in Spain found in 2024 that children born without fingerprints — a very rare condition called **adermatoglyphia** — often have skin that is a little drier and less flexible than average.

⑥ Why are everyone's fingerprints so different? It is not because our DNA writes every line of every print. A baby's fingerprints begin to form in the mother's body at about the thirteenth week of pregnancy, in a very delicate layer of skin. Tiny pushes from inside (bones, blood vessels) and outside (the position of the baby's hand, the movement of the baby, the flow of amniotic fluid) bend the skin in unique ways. Two babies with the same DNA can still have different prints, because their hands move and float in slightly different ways. Once the print is set, it will not change for the rest of the person's life — even if you cut or burn the surface, the same print will grow back, because the pattern is kept deeper inside the skin.

⑦ This is why fingerprints are so useful for police work and safe unlocking. A thumb on a phone screen has to match a pattern that is unique in the world. It is also why long ago, in ancient China, people used inked fingerprints on important documents. A signature can be copied, but a fingerprint cannot. That ancient idea, passed down over many hundreds of years, is the same one now on every modern smart phone in your pocket.

⑧ So the small loops and whorls on our fingertips are not a small accident. They are a careful little tool — helpful for wet surfaces, helpful for touch, helpful for the long, quiet care of our own skin. They are also a quiet reminder that each of us carries, on our own hands, a design that no other person on Earth will ever share.

根据文章内容，选择最佳答案。

1. What did the 2009 Manchester experiment find about fingerprints and grip?

A. Ridged fingers gripped much better than smooth ones.
B. Flat, smooth fingers gripped just as well as ridged ones, and sometimes a little better, so "grip" is only part of the story.
C. Fingerprints reduce friction to zero.
D. Smooth fingers cannot hold anything at all.

2. Why do fingerprints help us handle wet surfaces, according to the passage?

A. Because they attract more water.
B. Because the valleys between the ridges let extra water escape, like the grooves on a car tyre in rain.
C. Because the ridges become sticky when wet.
D. Because they completely stop water from touching the skin.

3. How might fingerprints make our fingers more sensitive?

A. The ridges shake at a certain speed against bumps on a surface, sending richer signals to the brain.
B. The ridges make the brain feel nothing.

C. They keep the finger warm.
D. They change colour when touching an object.

4. Why are everyone's fingerprints different, even identical twins'?

A. Because parents decide them at birth.
B. Because DNA writes every line of the print.
C. Because tiny inside and outside pushes bend the delicate skin in unique ways while the baby is still in the mother's body.
D. Because the baby chooses a pattern it likes.

5. What is the main idea of the passage?

A. Fingerprints are only used by the police.
B. Fingerprints serve several quiet jobs — helping with wet surfaces, touch, and skin care — and every person's pattern is unique because of tiny movements before birth.
C. Fingerprints disappear after age 20.
D. Identical twins share the same fingerprints.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "flat, smooth fingers gripped just as well..."
【解析】光滑手指握力相当甚至更好。A/C/D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "valleys between them... like the grooves on a car tyre..."
【解析】沟槽排水保持接触。A/C/D 均错。故选 B。

3.
【答案】A
【题型】细节理解题
【定位】第④段 "the ridges shake at a certain speed..."
【解析】振动传递触觉信号。B/C/D 均错。故选 A。

4.
【答案】C
【题型】推理判断题
【定位】第⑥段 "Tiny pushes from inside... outside... bend the skin in unique ways."
【解析】发育过程中的微小外力影响指纹。A/B/D 均错。故选 C。

5.
【答案】B
【题型】主旨大意题
【定位】通篇与第⑧段
【解析】多功能且唯一。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **ridge** [n.] 脊、棱
- **whorl** [n.] 螺旋纹
- **friction** [n.] 摩擦
- **amniotic fluid** 羊水
- **adermatoglyphia** [n.] 无指纹症